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Stokes-Based 3D Polarimetric Analysis of Dipolar Near-Fields

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ABSTRACT

Characterizing and manipulating light polarization at the nanoscale is increasingly important in advancing nanophotonic technologies, particularly as photonic components shrink to subwavelength dimensions. Dipole emitters, which effectively model many nanoscale light sources such as quantum dots and single molecules, play a central role in understanding light–matter interactions. While their radiation patterns are well-established, their polarization behavior remains nontrivial due to the complex interplay between radiative and evanescent field components. This intrinsic polarization mixing complicates the analysis of emitted light, particularly across the near-to-far-field transition. In this work, the 3D Stokes formalism is used to systematically analyze the polarization properties of various dipole types, including linear, spinning, and chiral dipoles, from the near- to the far-field. Following the deterministic nature of dipolar fields, 3D polarization features are reduced to the local intrinsic 2D frame, allowing an accessible evaluation of polarimetric features with respect to energy propagation.

1 | Introduction

The comprehensive characterization of light polarization at the nanoscale is increasingly recognized as a critical driver in the advancement of nanophotonic technologies, prompting significant efforts to develop innovative techniques for this purpose [1, 2]. As photonic systems approach subwavelength dimensions, electromagnetic fields become inherently nonparaxial, exhibiting a lack of a preferential propagation direction [3]. Examples of this include evanescent waves in interfaces [4] or around plasmonic nanoparticles [5]. In such conditions, conventional polarimetric frameworks, grounded in transverse fields, lack the formal generality necessary to capture the fully 3D nature of these waves.

To address this issue, several formalisms for the description of 3D polarization have been proposed. Early efforts pioneered by Richards and Wolf [6, 7] were based on the Angular Spectrum Representation [8], while a 3D extension of the Stokes formalism has been popular during the last decades [9]. These theoretical advances have been equally tied with experimental measurements of these three-dimensional polarization states, for which nanoscale probes or far-field imaging techniques have been recently developed [10–13]. Alternatively, dipolar approximations in lossless and axially symmetric scatterers allow reconstruction of near-field structure from well-controlled far-field polarimetric measurements [14].

Within this context, dipole emitters, as analytical surrogates of a wide range of nanoscale light sources and scatterers [15–19], have received significant attention for nanophotonic applications. While the coupling of single dipole emitters to nanoantennas has been extensively studied for different applications like directional emission [20, 21], fluorescence enhancement [22, 23], or improved efficiency in light-emitting diodes [24]; their polarization continues to exhibit surprising behaviors. Notably, Neugebauer et al. [25] and Joos et al. [26] demonstrated, independently, that a purely linear dipole can emit circularly polarized light.

Although the coupling of dipolar emitters to such systems can be directly calculated, the complexity of the resulting mathematical structure has favored the interpretation of these and other phenomena, including spin-momentum locking with spinning dipoles, from the inherent polarization properties of the emitters themselves. In particular, the emission of circular light from linear dipoles is understood from a mixture of TE/TM modes from evanescent components of the wave, a phenomenon also observed in total internal reflection [27] that results in a nonzero transverse spin; while advanced spin-momentum locking phenomena are typically linked to an engineered spin in the combination of different dipole moments [28]. These results evidence the major role of the inherent polarization of dipolar emitters in the development of polarization-dependent devices and mechanisms.

However, the description of these inherent polarization features has traditionally been done in terms of angular spectrum or spin formalisms, and not within a full polarimetric framework based on observables. With experimental achievements in the precise positioning of dipole emitters in complex nanophotonic systems [29–32], as well as in the experimental nonparaxial polarimetry of dipolar fields [13], a full polarimetric diagnostic tool for deterministic dipole emitters could guide future efforts in the development of novel nanoscale devices or techniques, including improvements in tip-enhanced spectroscopies [33] or Förster resonance energy transfer optimizations [34].

In this work, we employ the 3D Stokes formalism to systematically analyze the polarization states emitted from the near- to the far-field by various types of dipoles (i.e., linear electric and magnetic dipoles [16], spinning electric and magnetic dipoles [35], and chiral dipoles [36, 37]) excited by plane waves, offering an unified framework for comparing their emission polarimetric characteristics. This formalism offers a compact and algebraically straightforward method for determining the polarization state of the emitted radiation, expressed directly through the well-established Stokes parameters. In contrast to the vectorial angular spectrum method [8] used by other authors [25], our Stokes approach allows for a local, real-space phenomenological characterization of all polarization features that directly connects to potential nanoscale applications. Furthermore, we show how the local 3D polarization information can be compressed into a 2D ellipse, following the deterministic nature of dipolar fields. This step enables the straightforward interpretation of the local polarimetric information into a 2D formalism as well as facilitating the diagnosis of spin features considering propagating and reactive energy considerations.

2 | Theoretical Background

In the context of nanophotonics, in-plane dipolar emissions are a good approximation of light scattering by small plasmonic and dielectric particles [25, 26, 28, 38–41]. A typical scattering geometry for these measurements, as depicted in Figure 1a, consists in these particles lying in the XY plane, with light shone along the z direction with finite NA objectives.

The emission of such dipoles, in the far-field limit, is characterized by fields that are transverse to the Poynting vector, which is oriented in the radial direction. Thus, polarizations in the far-field limits can be generally evaluated in the usual 2D Stokes formalism using spherical coordinates. However, the field in the near- and intermediate-field regimes will have a nontrivial mixture of radial and transverse components. Due to the deterministic nature of dipolar fields, the field at each point can be locally described via a polarization ellipse lying in a 2D plane. However, the spatial inhomogeneity of the fields implies that this plane may vary significantly from point to point and a global 2D description cannot be achieved. In this context, the 3D Stokes

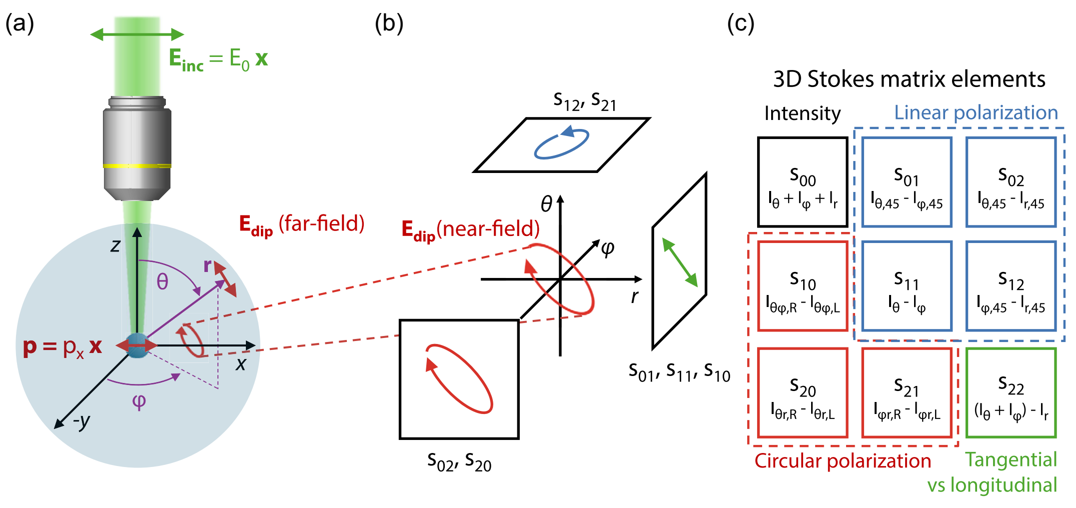

FIGURE 1 | (a) Scattering geometry considered in this work. A small nanoparticle can be modeled as a point dipole whose polarizability matches the polarization direction of the incident light (in this case, a linearly polarized wave in the x direction). (b) An arbitrary polarization ellipse from the dipole's near-field is decomposed into three spherical projections, each one with its pertaining Stokes parameters. (c) The different Stokes parameters, apart from the intensity term s_{00}, can be attributed to linear or circular polarization characteristics, or to the tangential/longitudinal character of the wave (s_{22}). The 45 subindex refers to a 45 degree-rotation direction from the spherical basis vectors (r, θ or φ), and the R, L indices represent right- or left- circular bases around their respective spherical basis vectors.

formalism can provide a clearer description. This formalism extends from the usual Stokes formalism, limited to 2D light fields, with a 3D coherency matrix [9]:

$$\overline{\mathbf{R_{3D}}} = \langle \mathbf{E_{3D}} \otimes \mathbf{E_{3D}}^{\dagger} \rangle = \begin{pmatrix} E_{3D,1}E^{*}_{3D,1} & E_{3D,1}E^{*}_{3D,2} & E_{3D,1}E^{*}_{3D,3} \\ E_{3D,2}E^{*}_{3D,1} & E_{3D,2}E^{*}_{3D,2} & E_{3D,2}E^{*}_{3D,3} \\ E_{3D,3}E^{*}_{3D,1} & E_{3D,3}E^{*}_{3D,2} & E_{3D,3}E^{*}_{3D,3} \end{pmatrix} \tag{1}$$

where † defines the Hermitian conjugate, the Kronecker product, and * the complex conjugate. The spatial and temporal dependence of fields has been omitted for the sake of brevity, and the vector components are denoted as the 1, 2, and 3 subindices. Due to the deterministic nature of dipolar fields, the time average has been omitted. This 3D coherency matrix can be expanded in terms of the Gell–Mann matrices, allowing to obtain the 3D Stokes parameters from its elements r_{ij}:

$$\begin{array}{lll} s_{00}=r_{11}+r_{22}+r_{33} & s_{01}=r_{12}+r_{21} & s_{02}=r_{13}+r_{31} \\ s_{10}=i(r_{12}-r_{21}) & s_{11}=r_{11}-r_{22} & s_{12}=r_{23}+r_{32} \\ s_{20}=i(r_{13}-r_{31}) & s_{21}=i(r_{23}-r_{32}) & s_{22}=\dfrac{r_{11}+r_{22}-2r_{33}}{\sqrt{3}} \end{array} \tag{2}$$

The interpretation of the 3D Stokes parameters can be built by analogy with the 2D parameters, considering each of the ellipse's projections to the characteristic planes of the chosen coordinate system (Figure 1b). For instance, s_{01}, s_{02}, and s_{12} are analogous to the 2D s_2 parameter (linear polarization difference between 45°-rotated components) in the planes defined by the (1, 2), (1, 3), and (2, 3) component pairs, while s_{11} is the 3D counterpart to the s_1 parameter (linear polarization difference between components). A similar connection can be traced between the 2D circular polarization parameter s_3 and the 3D s_{10}, s_{20}, and s_{21} parameters.

From this analogy, two subsets of parameters appear, characterizing the linear and circular polarization content of the ellipse's projections, and s_{00} functions as a measure of the total intensity, as s_0 does in 2D polarization states.

The final 3D Stokes parameter (s_{22}) measures the projection strength of the third component against the other two, with no obvious counterpart in the 2D formalism. As the radial direction of the electric field is privileged (vanishing as the far-field limit is approached), we consider an alternative spherical coordinate system (θ, φ, r), which conserves the handedness of the traditional spherical coordinates and places the privileged $\boldsymbol{r}$ direction as the third component of the field. This allows s_{22} to measure the apparent transversality of the fields, as well as separating the distinct near- and far-field regimes. Figure 1c summarizes the interpretation of the 3D Stokes parameters: considering the convenient alternative spherical coordinate system mentioned earlier, each element can be equivalently described in terms of the intensity differences (e.g., $s_{11} = I_\theta - I_\varphi$).

3 | Results and Discussion

Here, we test the 3D Stokes formalism as an unified framework to systematically analyze the polarization states of various dipoles. We start with the linear electric dipole, which has received significant attention, before analyzing the spinning electric dipole as an extension of it. Next, the linear and spinning magnetic dipoles are analyzed; and finally, a chiral dipole—a combination of both electric and magnetic dipoles—is studied.

3.1 | Linear Electric Dipole

We begin our discussion by considering a linear electric dipole, previously studied in [25] and [26]. We employ the well-known expression for the electric field of the electric dipole $\boldsymbol{E}_\mathbf{p}$:

$$\boldsymbol{E}_\mathbf{p} = \frac{e^{ikr}}{4\pi\varepsilon r}\left[\mathbf{p}\left(k^2+\frac{ik}{r}-\frac{1}{r^2}\right)-\mathbf{n}(\mathbf{n}\cdot\mathbf{p})\left(k^2+\frac{3ik}{r}-\frac{3}{r^2}\right)\right] \tag{3}$$

where the intrinsic regimes of near-, intermediate-, and far-field are encoded in the r^{-3}, r^{-2}, and r^{-1} scaling terms, respectively. Setting $\mathbf{n}$ in the radial direction $\mathbf{n}=(0,\ 0,\ 1)^T$ in the modified spherical coordinate system) and with the dipole direction as $\mathbf{p}=p_x\mathbf{u_x}$, $\boldsymbol{E}_\mathbf{p}$ is given as:

$$\boldsymbol{E}_\mathbf{p} = p_x\frac{e^{ikr}}{4\pi\varepsilon r^3}\begin{pmatrix} [-1+kr(i+kr)]\cos\theta\cos\varphi \\ [1-kr(i+kr)]\sin\varphi \\ 2(1-ikr)\sin\theta\cos\varphi \end{pmatrix}\begin{pmatrix} u_\theta \\ u_\varphi \\ u_r \end{pmatrix} \tag{4}$$

The analytical nature of the dipole field allows us to derive closed-form expressions for the 3D Stokes parameters, which are provided in the Supporting Information (Section S1). To show the basic polarimetric behavior of the electric dipole field, we plot in Figure 2 the corresponding 3D Stokes parameters at different-distance spherical surfaces enclosing the dipole. The different radial distances correspond to the near-field ($r=\frac{\lambda}{50}$), intermediate ($r=\frac{\lambda}{3}$) and far-field ($r=2\lambda$).

The 3D Stokes formalism allows to see the fully linear polarization in the near-field regime (Figure 2a), as the three Stokes parameters related to circular polarization (s_{10}, s_{20} and s_{21}) are all zero. The vectorial character of this linear polarization is further revealed by analyzing the dominant components for significant points. In particular, since s_{22} maps the transversality of the field with respect to the radial direction, we can see that the field is dominated by the radial component at the intensity maxima (located around $\theta=\pi/2$ and $\varphi=[-\pi,\ 0,\ \pi]$). As one moves away from these points, the 3D parameters also pinpoint the rotation of the fields. For example, the negative s_{11} reveals the field becoming azimuthal for $\varphi \sim\ \pm\pi/2$.

The features observed in the transition from the near- field to the far-field are also captured by the formalism. For the intermediate regime (Figure 2b, at $r=0.375\lambda$), while the s_{01}, s_{11}, and s_{22} show similar profiles, nonzero circular parameters s_{20} and s_{21} appear. These correspond to components of the electric field spin $\mathbf{s_E} \propto \Im\mathfrak{m}(\mathbf{E}^*\times\mathbf{E})$, in either the azimuthal or zenithal directions, and transverse to the propagation direction. These are the real-space equivalents of the transverse spin observed by Neugebauer et al. [25] and exploited by Joos et al. [26] for polarization control in optical fiber excitation. As opposed to the angular-spectrum treatment, this transverse spin appears in real space as a result of the interference of radiative (r^{-1}) and reactive (r^{-2} and (r^{-3})) contributions with non-negligible longitudinal components.

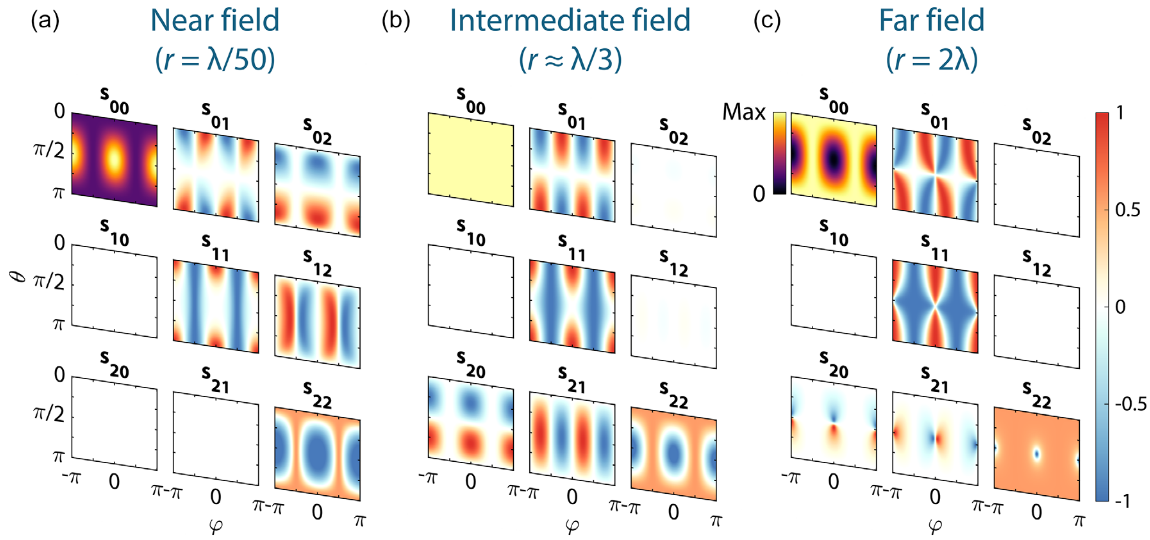

FIGURE 2 | Normalized 3D Stokes parameters distribution over spherical surfaces enclosing the linear electric dipole $\mathbf{p} = p_x \mathbf{u_x}$ and radii corresponding to (a) the near-field ($r = \lambda/50$), (b) the intermediate-field ($r \approx \lambda/3$), and (c) the far-field ($r = 2\lambda$). All parameters are normalized by s_{00}, with the exception of s_{00} itself, which denotes the total electric field intensity.

Interestingly, the intensity profile of the electric dipole (s_{00}) is uniform at this distance, meaning that the ellipticity of these fields is not a product of numerical artifacts but an actual physical phenomenon.

In contrast to this, the far-field regime (Figure 2c) displays non-zero radial parameters (s_{20} and s_{21}) only at points where radiation is suppressed ($s_{00} \sim 0$), pointing to the total transversality of the field as the far-field limit is approached. This is very clearly seen in the s_{22} parameter map, displaying an uniform $1/\sqrt{3}$ value at all points except for the intensity minima.

Although powerful, the interpretation of the 3D Stokes parameters for such inhomogeneous fields is not trivial, in contrast with the relatively straightforward interpretation of 2D parameters. To enhance the comprehension of the relationship between the 2D and 3D parameters, we exploit the cylindrical symmetry of the linear electric dipole and analyze the field at $\varphi = 0$:

$$\boldsymbol{E}_{\mathbf{p}}(\varphi = 0) = p_x \frac{e^{ikr}}{4\pi\varepsilon r^3} \begin{pmatrix} (-1 + ikr + k^2r^2)\cos\theta \\ 0 \\ 2(1 - ikr)\sin\theta \end{pmatrix} \begin{pmatrix} u_\theta \\ u_\varphi \\ u_r \end{pmatrix} \quad (5)$$

At this plane, the field can be 2D, containing only radial and zenithal components and thus allowing a conventional 2D Stokes treatment where $s_0 = s_{00}$, $s_1 \simeq s_{22}$, $s_2 = s_{02}$, and $s_3 = s_{20}$. From the 2D Stokes parameters (given in the Supporting Information, Section S2), degrees of circular and linear polarization can be obtained as $\mathrm{DOCP_{2D}} = \frac{s_3}{s_0}$ and $\mathrm{DOLP_{2D}} = \sqrt{1 - \mathrm{DOCP_{2D}^2}}$. The computation of these scalar quantities allows a simple representation of the overall polarization distribution across the $\varphi = 0$ plane, which is provided in Figure 3a.

The main features of Figure 2 are replicated here, with the field being predominantly linear in the near- and far-field limits, and a near-unity degree of circular polarization in the intermediate

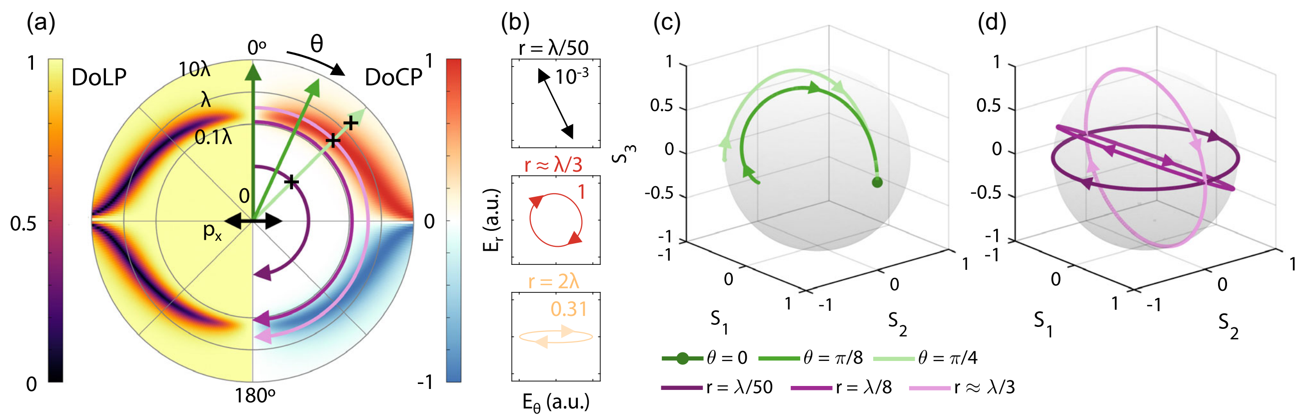

FIGURE 3 | Polarization features of a linear electric dipole in the $\varphi = 0$ plane. (a) Distribution of DOLP and DOCP. (b) Polarization ellipses of the electric field at the crossed points in (a), with their respective DOCP values. (c) Evolution of the polarization over the green radial trajectories in (a) over the Poincaré sphere. (d) Evolution of the polarization over the purple zenithal trajectories in (a) over the Poincaré sphere.

regime for $\theta \neq \{0, \pm\pi/2, \pi\}$. As θ evolves toward $\pm\pi/2$, this regime of circular polarization shifts asymptotically to the far-field, where radiation is suppressed. The simplified structure of the field in Equation (5) evidences the origin of this elliptical polarization as a mixture of the $(kr)^2$ term in the zenithal component and the $-ikr$ term in the radial field. As a result, elliptical polarizations appear only in the near-intermediate zones of the field, where these terms have similar magnitudes.

As the regular Stokes parameters are obtained, a polarization ellipse can be traced at any point for space. In particular, for $\theta = \pi/4$, the evolution of the fields becomes quite clear, as depicted in the panels of Figure 3b: The near-field has a purely linear field ($\mathrm{DOCP_{2D}} \sim 10^{-3}$), whose dominant component is the radial. As distance grows, the field becomes circular ($\mathrm{DOCP_{2D}} \sim 1$ for $r \approx \lambda/3$) and, as $r$further increases, becomes linear again, with the radial component approaching zero. However, it should be noted that for relatively large distances ($r = 2\lambda$), the field can still retain a significant degree of circular polarization ($\mathrm{DOCP_{2D}} \approx 0.3$).

To obtain further information on the individual Stokes parameters, we describe trajectories over the $\varphi = 0$ plane as trajectories over the Poincaré sphere surface. Again, such a representation is only possible with two-dimensional polarization states, although 3D extensions such as the Hannay–Majorana and Poincarana constructions exist [42, 43]. Such constructions, however, have relatively complex interpretations, unlike the regular Poincaré sphere.

Focusing on near- to far-field transitions (radial trajectories, depicted by green lines), both the near- and far-field limits are characterized by equatorial locations, referring to purely linearly polarized states. However, the zenithal coordinate θ at the near-field determines the specific properties of the linear state: as seen in Figure 3d, which depicts the constant-radius angular trajectories (pink arrows) in Figure 3a, a $[0, \pi]$ angular trajectory produces a full lap around the Poincaré sphere equator. This representation predicts the results found by Joos et al. [26], where the orientation of linearly polarized light coupled to optical fibers is controlled by the position of a nearby nanorod, acting as a linear dipolar source.

For longer distances, angular trajectories become slightly tilted until $r \approx \lambda/3$, where the laps become meridional, as points here approach the sphere poles, i.e., fully circular polarizations. For longer distances, as the far-field limit is approached, radial trajectories collapse into a single point where $s_1/s_0 = 1$, corresponding to the full zenithal polarization of such far-field radiation. In particular, $\theta = 0$, corresponding to the emission maximum, has a fixed polarization along that same point for all radial distances. For longer distances, the angular trajectories remain similar, which represents well the idea that the regime of circular polarization shifts asymptotically to the far-field as θ approaches $\pi/2$. Again, it should be noted that this situation remains unphysical due to the far-field suppression at those angles.

3.2 | Spinning Electric Dipole

For dipolar scatterers or emitters, excitation by a linearly polarized electromagnetic wave induces an electric dipole moment that oscillates linearly along the polarization direction of the incident field. Conversely, circularly polarized excitation can be also considered, resulting in spinning dipolar moments. For an electric dipole, this spinning moment can be modeled as $\mathbf{p} = p_x\mathbf{u_x} + ip_y\mathbf{u_y}$. Using this definition, the electric field becomes:

$$\mathbf{E_{p,CP}} = p\frac{e^{ikr}}{4\pi\varepsilon r^3}(\cos\varphi + i\sin\varphi)\begin{pmatrix}[-1+kr(i+kr)]\cos\theta \\ i[-1+kr(i+kr)] \\ 2(1-ikr)\sin\theta\end{pmatrix}\begin{pmatrix}u_\theta \\ u_\varphi \\ u_r\end{pmatrix} \tag{6}$$

where we have set $p_x = p_y = p$ to match a perfectly circular excitation. Different values of p_x and p_y, as well as modifications of their relative phase, can be used to model excitation by an elliptically polarized wave. Again, closed-form expressions for the 3D Stokes parameters can be derived, and are provided in the Supporting Information (Section S1). The 3D Stokes parameters are plotted in Figure 4.

In general, the presence of two orthogonal dipoles produces polarization patterns that are uniform in φ, resembling an averaging of the values of the linear dipole. As such, parameters with sign changes over φ in the linear dipole (s_{01}, for instance) often result in zero values for the spinning dipole. Likewise, uniform zero values of parameters in the linear dipole often result in nonzero values for the spinning. A clear example of this is found in the circular polarization element s_{10}, where the 3D Stokes formalism clearly reflects the well-known helicity change [35] at $\theta = \pi/2$ for all field regimes.

Another interesting feature displayed by the Stokes parameters is the presence of a zenithal uniform-helicity circular polarization (s_{21}) in the near-field regime (Figure 4a), which also matches the intensity maximum ($\theta = \pi/2$). This result matches the transverse spin-momentum locking phenomena described by [28, 44–46], where the transverse spin orientation is intrinsically linked to the propagation direction due to field confinement and longitudinal components. As distance grows (Figure 4b), this circular polarization component vanishes, while the characteristic transverse spin of linear dipoles (s_{20}) manifests. Interestingly, the spinning dipole showcases an uniform mixture of radial and azimuthal polarizations at $\theta = \pi/2$ (s_{12}), which correspond to the local linear states of some vortex beams [47, 48].

As the far-field limit is approached (Figure 4c), the radial fields are again lost ($s_{22} = 1/\sqrt{3}$), recovering a 2D representation of the field. Here, the main polarization component corresponds to the well known opposite circular polarizations in the radial direction (given by s_{10}). Following the results of the intermediate-field, at $\theta = \pi/2$ the field is azimuthally polarized, as given by the dominant negative s_{11}. It is worthy to discuss that these polarization results in the intermediate- and far-field could lead to the coupling of vortex modes in nearby systems [49].

3.3 | Magnetic Dipoles: Linear and Spinning

A magnetic dipole $\mathbf{m} = m_x\mathbf{u_x}$ has a simpler electric field expression:

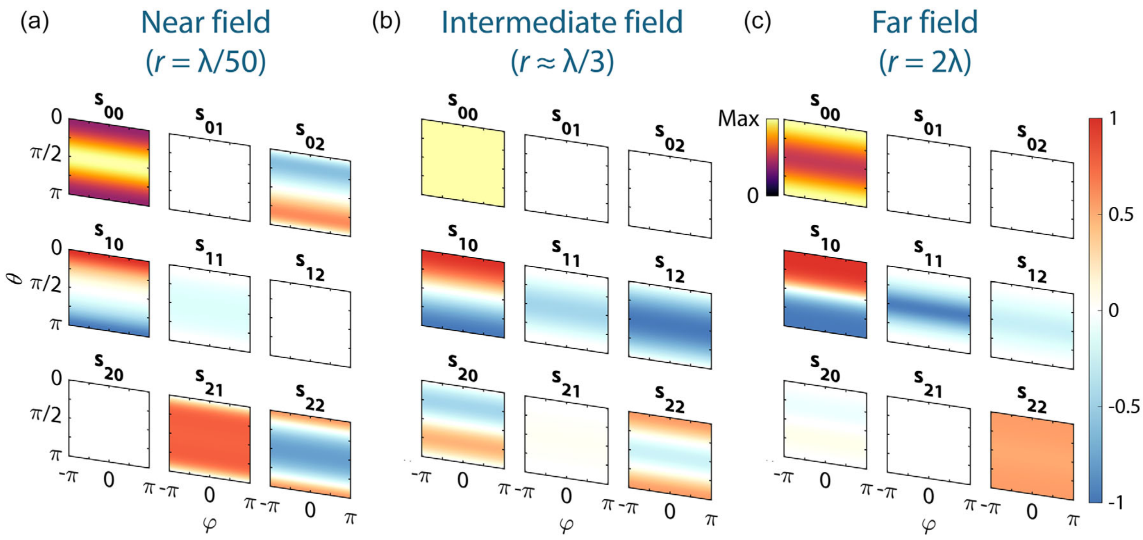

FIGURE 4 | Normalized 3D Stokes parameters distribution over spherical surfaces enclosing the spinning electric dipole $\mathbf{p} = p_x\mathbf{u_x} + ip_y\mathbf{u_y}$ and radii corresponding to (a) the near-field ($r = \lambda/50$), (b) the intermediate-field ($r \approx \lambda/3$), and (c) the far-field ($r = 2\lambda$). All parameters are normalized by s_{00}, with the exception of s_{00} itself, which denotes the total electric field intensity.

$$\boldsymbol{E}_\mathbf{m} = -m_x \frac{e^{ikr}}{4\pi c\varepsilon r^2} k(i + kr) \begin{pmatrix} \sin\varphi \\ \cos\theta\cos\varphi \\ 0 \end{pmatrix} \begin{pmatrix} u_\theta \\ u_\varphi \\ u_r \end{pmatrix} \tag{7}$$

where it becomes clear that the field has is always transverse to the radial direction and the conventional 2D Stokes treatment can be applied (with $s_0 = s_{00}$, $s_1 = s_{11}$, $s_2 = s_{01}$ and $s_3 = s_{10}$). It is shown, however, that $s_3 = 0$ and thus a linear magnetic dipole emits linearly polarized light only. Furthermore, the expressions for the 2D Stokes parameters and a full description of its features are given in Section S1.

A similar treatment can be applied to a magnetic dipole, with a spinning moment $\mathbf{m} = m_y\mathbf{u_y} - im_x\mathbf{u_x}$, which assures the same helicity as the spinning electric dipole. The field emitted by such a dipole is:

$$\boldsymbol{E}_{\mathbf{m,CP}} = m \frac{e^{ikr}}{4\pi c\varepsilon r^2} k(i + kr) \begin{pmatrix} \cos\varphi + i\sin\varphi \\ \cos\theta(\cos\varphi + i\sin\varphi) \\ 0 \end{pmatrix} \begin{pmatrix} u_\theta \\ u_\varphi \\ u_r \end{pmatrix} \tag{8}$$

where, again, $m_x = m_y = m$. Again, and as expected, the field has only two nonzero components with the same radial evolution, meaning that a 2D Stokes conventional formalism at a single distance can provide the full polarimetric picture of this dipole's emission. In contrast to the linear magnetic dipole, s_2 vanishes while the remaining Stokes parameters are non-zero, resulting in a richer polarimetric structure. The Stokes parameters expressions and their values over space are provided in the Supporting Information (Section S1). In a similar fashion to electric spinning dipoles, magnetic dipoles are characterized by zenithal linearly polarized states at $\theta = \pi/2$.

3.4 | Chiral Dipole

The final case we analyze in this work corresponds to a chiral dipole $\mathbf{p}_\xi$, usually employed to model fluorescent chiral molecules or nanoparticles. This dipole can be understood as a collinear combination of electric and magnetic dipoles:

$$\mathbf{p}_\xi = (p + ic\xi m)\mathbf{u_p} \tag{9}$$

with c being the speed of light and ξ the Pasteur parameter linking the strength of electric and magnetic dipoles, whose norm is considered equal. This parameter has an upper limit value of 1 and typically has small values [50, 51]. Considering linear dipoles $\mathbf{p} = p_x\mathbf{u_x}$ and $\mathbf{m} = m_x\mathbf{u_x}$, the total field is given by:

$$\boldsymbol{E}_\xi = \frac{e^{ikr}}{4\pi c\varepsilon r^3} \begin{pmatrix} cp_x[-1 + kr(i + kr)]\cos\theta\cos\varphi + \xi k m_x r(1 - ikr)\sin\varphi \\ cp_x\sin\varphi + kr(1 - ikr)(\xi m_x\cos\theta\cos\varphi - icp_x\sin\varphi) \\ 2cp_x(1 - ikr)\cos\varphi\sin\theta \end{pmatrix} \begin{pmatrix} u_\theta \\ u_\varphi \\ u_r \end{pmatrix} \tag{10}$$

Although optical chirality or helicity cannot be formally evaluated from pure electric field measurements [52], the 3D polarization framework here can illustrate polarimetric signatures associated with electric–magnetic coupling, which could be useful in Circularly Polarized Luminescence techniques [53]. With that purpose in mind, we analyze the maximal chirality case $\xi = 1$ in Figure 5. A more realistic case ($\xi < 1$) is examined in the Supporting Information (Section S3).

As the chiral dipole results from the addition of collinear electric and magnetic dipoles, with electric dipoles having stronger near-fields (owing to the r^{-3} term as opposed to the r^{-2} magnetic dipole contribution), the near-field limit of a chiral dipole (Figure 5a) resembles very closely the linear electric dipole.

Magnetic dipole contributions become more relevant at the intermediate regime (Figure 5b), and the 3D formalism allows to distinguish several differences. First, magnetic contributions break the uniform intensity found in other dipoles, with a closer

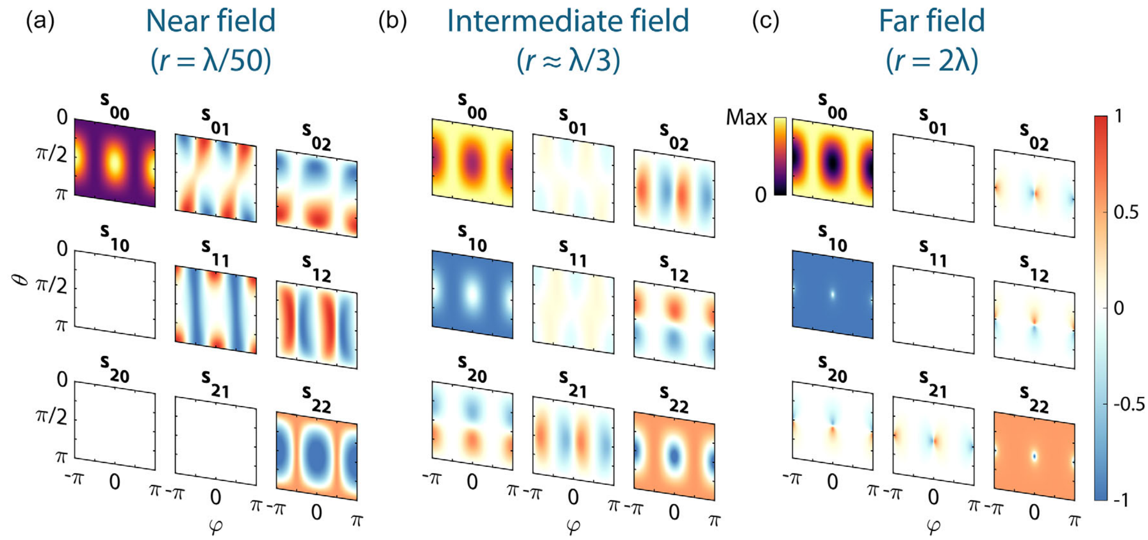

FIGURE 5 | Normalized 3D Stokes parameters distribution over spherical surfaces enclosing the chiral dipole $\mathbf{p}_\xi = p_x + icm_x\mathbf{u_x}$ (taking $\xi = 1$) and radii corresponding to (a) the near-field ($r = \lambda/50$), (b) the intermediate-field ($r \approx \lambda/3$), and (c) the far-field ($r = 2\lambda$). All parameters are normalized by s_{00}, with the exception of s_{00} itself, which denotes the total electric field intensity.

resemblance to the usual far-field pattern. The magnetic contributions are also evidenced by circular polarization in the radial direction (s_{10}) in these maximum intensity areas. In contrast to the spinning electric dipole, the s_{10} parameter has an uniform sign, a clear distinction between spinning and chiral dipoles. Apart from s_{10}, the Stokes treatment reveals other circular polarization parameters (s_{20} and s_{21}) with a similar shape to the electric dipole, meaning that the chiral dipole should couple circular polarization to waveguides and other structures much like the linear electric dipole. As these polarizations are not related to the chiral parameter ξ, these circular polarization can become undesired polarization artifacts in chiral sensing applications, particularly for small values of ξ.

Finally, the far-field (Figure 5c), much like the electric dipole, presents a fully transversal field ($s_{22} = 1/\sqrt{3}$)—in this case, fully circularly polarized in the radial direction (s_{10}).

Additionally, and similarly to the linear electric dipole, cylindrical symmetry can be applied to characterize the Stokes parameters over the $\varphi = 0$ plane. However, the field remains three-dimensional at this plane, meaning that the full 3D Stokes formalism must still be applied. The expressions for the field and the Stokes parameters at $\varphi = 0$ are given in Section S2 for further comparison.

3.5 | Tracing the Embedded 2D Polarization States

Despite the ability of the 3D Stokes formalism to rigorously describe the global polarimetric landscape of dipolar emission, the mathematical form of the Stokes parameters is strongly dependent on the choice of reference system, which obstructs the physical interpretation of the local polarization state.

As stated earlier, the deterministic nature of coherent dipolar fields prevents them from having true 3D polarization states. This means that, for any point in space with a non-zero electric-field spin $\mathbf{s_E} \propto \Im\mathfrak{m}(\mathbf{E}^* \times \mathbf{E})$, a local 2D ellipse can be traced around it, defining an intrinsic coordinate system where the field takes its simplest and most physically transparent form.

Even though the spatially inhomogeneous dipolar fields make this intrinsic system change for each point in space, it is nevertheless useful to consider it to remove coordinate-dependent mixing between field components in laboratory systems and to provide a clearer picture when analyzing local polarization states. Details on the mathematical procedure to obtain the intrinsic system can be found in Section S4, and a MATLAB implementation is available at [54].

Furthermore, the intrinsic representation also helps to visualize the direction of local energy flow, given by the real part of the time-averaged Poynting vector $\mathbf{S} = \mathbf{E} \times \mathbf{H}^*$. In general, dipolar near-fields do not require orthogonality between electric and magnetic fields and the local energy flow, which means that similar polarimetric features might correspond to essentially different physical situations. Due to this, the electric spin vector and the energy flux vector may not be collinear, and will be separated by an angle θ_C:

$$\cos\theta_C = \frac{\mathbf{s_E} \cdot \Re\mathfrak{e}(\mathbf{S})}{|\mathbf{s_E}||\Re\mathfrak{e}(\mathbf{S})|} \tag{11}$$

This angle allows to introduce a spin transversality index τ_S as:

$$\tau_S = \sin\theta_C = \sqrt{1 - \left(\frac{\mathbf{s_E} \cdot \Re\mathfrak{e}(\mathbf{S})}{|\mathbf{s_E}||\Re\mathfrak{e}(\mathbf{S})|}\right)^2} \tag{12}$$

This quantity varies between 0 and 1, corresponding to perfect collinearity or transversality between spin and energy flow. Although τ_S is undefined when $|\mathbf{s_E}| = 0$, zero-spin fields are only found in linear magnetic dipoles, with trivial reductions to their intrinsic systems, and the strict near- and far-field limits of the other dipoles. Thus, these situations can be safely excluded from our analysis or extended from other results by continuity.

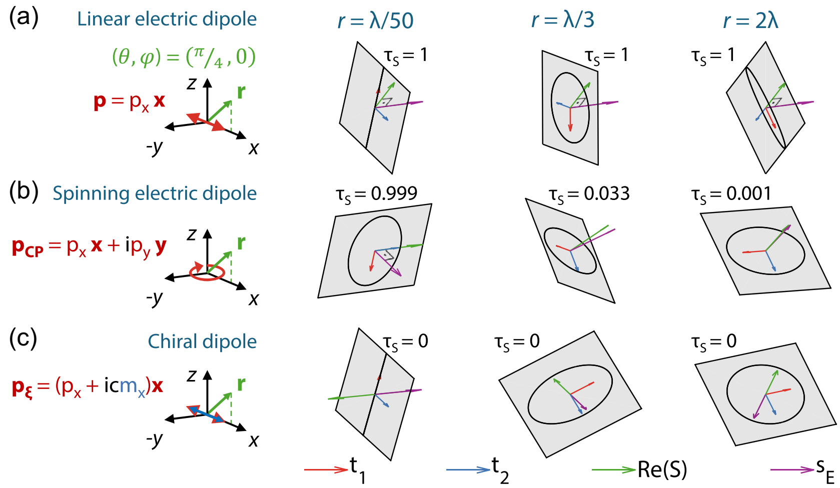

FIGURE 6 | 2D polarization ellipses (black) of dipoles in their intrinsic coordinate system (red and blue arrows) for (a) linear electric dipole $\mathbf{p} = p_x\mathbf{u_x}$, (b) spinning electric dipole $\mathbf{p_{CP}} = p_x\mathbf{u_x} + ip_y\mathbf{u_y}$, and (c) chiral dipole $\mathbf{p}_\xi = p_x + icm_x\mathbf{u_x}$ (taking $\xi = 1$). Polarization states are evaluated at the single direction $(\theta, \varphi) = (\frac{\pi}{4}, 0)$and radii corresponding to the near-field ($r = \lambda/50$), the intermediate-field ($r \approx \lambda/3$) and the far-field ($r = 2\lambda$) regimes. The spin is represented by the purple arrow and the real part of the Poynting vector by the green arrow.

Figure 6 shows the 2D ellipses of different dipolar emitters (linear electric, spinning electric, chiral) in their respective intrinsic systems, together with their respective τ_S. To evidence differences between different dipoles, the intrinsic system was evaluated at near-, intermediate-, and far-field points in a single direction $(\theta, \varphi) = \left(\frac{\pi}{4}, 0\right)$ for all cases.

As the chosen direction lies in the $\varphi = 0$ plane, the linear electric dipole's (Figure 6a) ellipses lie in that plane as well, reproducing Figure 3b. The visual representation of the intrinsic system also points to the complete transversality of the linear dipole spin and energy flux for all regimes: while the energy flux points in the radial direction, contained in the tangential plane, the spin vector is perpendicular to that plane.

Matching with the 3D Stokes representation, the spinning electric dipole (Figure 6b) shows a notable ellipticity for all regimes. In particular, for the near-field limit, the spin vector and the real part of the Poynting vector direction are almost perpendicular, similar to the electric-field spin observed in the electric dipole. This situation changes as we move farther away from the dipole, with almost parallel spins and energy flows already in the intermediate-field.

Finally, the chiral dipole (Figure 6c) deviates heavily in the energy flow direction from the other two cases, owing to magnetic field dipole contributions. Strikingly, for $\xi = 1$, spin and energy flow are collinear for all points in space, showcasing the important difference between electric and chiral dipoles: While similar polarimetric features can appear in their Stokes representation (like the similar linear polarizations in their near-fields, or the nonzero s_{20} in their intermediate-fields), electric dipole spin is always transversal, and chiral dipole spin is always longitudinal. It should be noted that Figure 6 represents a single direction in space, but, as shown in Section S5, the spin transversality index has relatively homogeneous values across the complete solid angle.

These fundamental differences between the different dipoles can be, interestingly, related to their chirality, characterized by the optical chirality density $C = \frac{\omega\epsilon}{2}\Im\mathfrak{m}(\mathbf{E} \cdot \mathbf{B}^*)$ [36, 55]. As shown in Section S5, linear electric dipoles, exhibiting a zero-chirality in all space (as a result of a global transversality between their electric and magnetic fields), only have transverse optical spin ($\tau_S = 1$). In the case of circular dipoles, where chirality manifests as the far-field is approached, its near-field is dominated by r^{-3} terms in the electric field that are not present in its magnetic field. This asymmetry between near electric and magnetic fields is translated in a negligible chirality, which allows to draw a similar conclusion to the linear electric dipole.

In contrast to these two examples, a chiral dipole enforces symmetry between electric and magnetic dipoles, resulting in a maximum chirality for all space and a perfect alignment of optical spin with energy propagation ($\tau_S = 0$). While this relationship between chirality and electric-field spin is neither trivial nor general, particularly as magnetic-field spin contributions play a role in the intermediate field, it may be an useful indicator for the coupling of helical fields.

4 | Conclusion

This work establishes the three-dimensional Stokes formalism as an unified real-space framework for analyzing polarization in dipolar electromagnetic fields across the near-, intermediate-, and far-field regimes. By expressing dipolar emission in terms of 3D Stokes parameters, we demonstrated a complete description of polarization features based on observable intensities, completing angular-spectrum approaches [25] into a local and phenomenological interpretation that links directly to potential nanoscale applications.

The implementation of this framework to linear and spinning electric and magnetic dipoles, as well as to chiral dipoles, reveals systematic differences in their polarization organization that are not immediately apparent from field expressions or specific couplings alone.

In particular, the consideration of the intrinsic reference frame of each dipole, together with the spin-transversality index proposed here, demonstrated that the circular polarization shown in the intermediate-field regime of electric dipoles corresponds to a spin transverse to the local energy flow. In contrast, ideal chiral dipoles display a fundamentally different behavior in which spin and energy flow remain collinear throughout space, reflecting a helicity-dominated field structure. These distinctions, therefore, highlight that similar Stokes signatures may correspond to physically different mechanisms, emphasizing the importance of combining polarimetric analysis with spin-energy diagnostics. Further physical interpretation of these chirality-related observations may be obtained from the formal consideration of the 6×6 electromagnetic polarization matrix [52, 56], where the chiral information of the Poynting vector is encoded in the antisymmetric part of its antidiagonal 3×3 blocks.

Our framework is not limited to the single emitters analyzed here, but also to more complex dipolar systems, like the Huygens and Janus dipoles [28, 38]. Beyond single dipolar systems, the 3D Stokes framework can be readily applied to systems containing several dipolar emitters or scatterers, particularly to those presenting fixed geometries, such as oligomer nanoantennas [57] or metasurfaces [58]. In such systems, the total field, resulting from the coherent addition of all dipole components, must be computed first with either numerical or semi-analytical methods.

Furthermore, incoherent ensembles of weakly interacting dipoles with fixed-geometries, including arrangements of organic emitters, rare-earth dopings, or quantum dots, can be assessed in a non-normalized way by the trivial addition of their individual 3D Stokes vectors. In that sense, the framework presented here could be useful for technologies such as LEDs [59, 60], rare-earth emitter devices [61], or the study of polarization in the thermal emission of metasurfaces [62]. Although incoherence prevents achieving a 2D reduction, the intrinsic reference frame can still be obtained to analyze the relationship of spin with energy flow. Further analysis in these systems may be drawn from the consideration of phase effects in mutual coherency matrices [63].

In conclusion, the presented framework provides a practical diagnostic tool for interpreting polarization structure in near- and intermediate-field optics, as well as its relationship with power flow. Thus, we believe that the application of this framework will be useful to the study of polarization in incoherent systems as well as in the design and development of polarization-dependent devices.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are openly available in DiPolIn (DIpolar POLarization in INtrinsic reference frame) at https://zenodo.org/records/18682022.

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